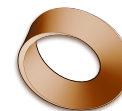


Name: _____

Date: _____

Mr. Rawson • Y9 MYP Physics



1.2 – Period and Frequency

Unit 1: Oscillations & Waves

Learning intentions

- a) I can define **period** as the time for one complete oscillation.
- b) I can define **frequency** as the number of oscillations per second.
- c) I can use the equation $f = 1/T$ to calculate frequency from period.
- d) I can convert between hertz, kilohertz, megahertz and gigahertz.

Theory

The **period** (T) is the time taken for one complete oscillation or wave cycle to occur. It is measured in seconds (s). For example, if a pendulum swings back to its starting position every 2 seconds, its period is 2 s. The **frequency** (f) is the number of complete oscillations that happen in one second. It is measured in hertz (Hz), where $1 \text{ Hz} = 1 \text{ oscillation per second}$.

Period and frequency are reciprocals of each other. This means they have an inverse relationship: as frequency increases, the period decreases. The mathematical relationship is defined by two equivalent equations: $f = 1/T$ $T = 1/f$

To use these equations correctly, units must be consistent. Frequency is usually in hertz (Hz). If given in kilohertz (kHz), megahertz (MHz) or gigahertz (GHz), you must convert to Hz first: - $1 \text{ kHz} = 1\,000 \text{ Hz}$ - $1 \text{ MHz} = 1\,000\,000 \text{ Hz}$ - $1 \text{ GHz} = 1\,000\,000\,000 \text{ Hz}$

Worked Example 1: Finding frequency from period A guitar string vibrates with a period of 0.002 s. What is its frequency? Step 1: Identify the known value. $T = 0.002 \text{ s}$. Step 2: Choose the correct equation. We need frequency, so use $f = 1/T$. Step 3: Substitute the value. $f = 1/0.002$. Step 4: Calculate. $f = 500 \text{ Hz}$.

Worked Example 2: Finding period from frequency (with conversion) A radio station broadcasts at a frequency of 98.5 MHz. What is the period of the wave in seconds? Step 1: Convert frequency to Hz. $f = 98.5 \times 1\,000\,000 = 98\,500\,000 \text{ Hz}$. Step 2: Choose the correct equation. We need period, so use $T = 1/f$. Step 3: Substitute the value. $T = 1/98\,500\,000$. Step 4: Calculate using standard form. $T \approx 1.02 \times 10^{-8} \text{ s}$.

Complete the sentence: The time taken for one complete oscillation is called the _____ and is measured in seconds.

Word bank: frequency · period · hertz · amplitude · wavelength

Activity

Measuring a Pendulum's Period

You will need: a clamp stand, a string, a bob (mass), a stopwatch.

- Set up a simple pendulum by attaching the string to the clamp and the bob to the other end. Ensure the length is about 50 cm.
- Pull the bob back slightly (less than 10 degrees) and release it without pushing.
- Start the stopwatch as the bob passes the lowest point moving in one direction. Count 10 complete oscillations (back and forth). Stop the watch exactly at the 10th pass.
- Record the total time for 10 oscillations in the table below. Repeat this twice to get three trials.
- Calculate the average time for 10 oscillations.

- Divide the average time by 10 to find the period (T) of one oscillation.

Trial	Time for 10 oscillations (s)
1	_____
2	_____
3	_____
Average time for 10	_____
Period (T)	_____

Questions

1. Define the term **frequency**.

2. A wave has a frequency of 50 Hz. Calculate its period in seconds.

3. An ultrasound wave has a period of 0.000002 s. What is its frequency in hertz?

4. Convert 3.5 MHz into hertz.

5. A sound wave has a period of 0.005 s. Calculate its frequency.

Extension

Research the typical frequency range of human hearing (20 Hz to 20 000 Hz). Calculate the longest and shortest periods for these frequencies. Explain why we cannot hear sounds with a period of 0.1 s.
